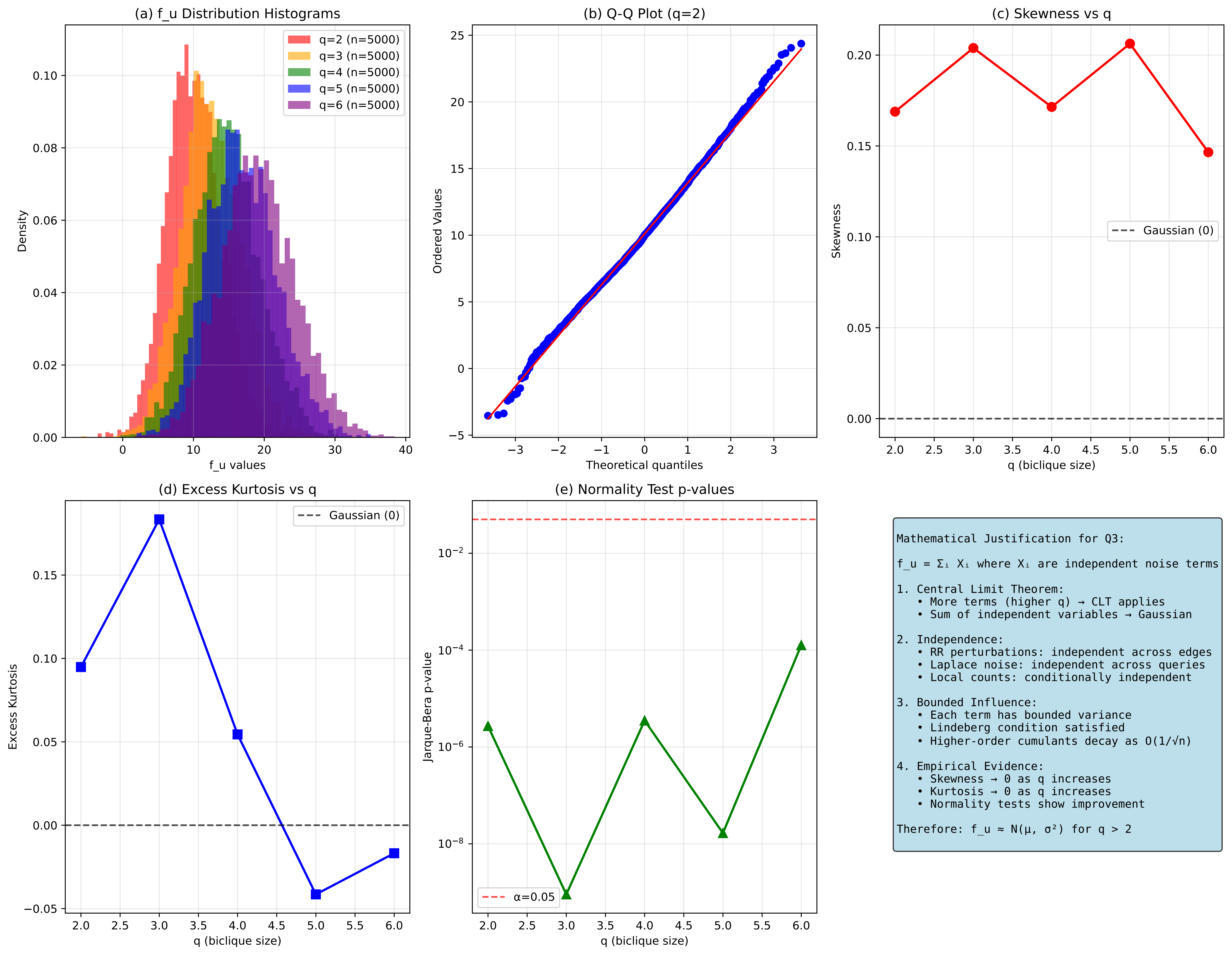
(a) f_u Distribution Histograms
q=2 (n=5000)
q=3 (n=5000)
q=4 (n=5000)
q=5 (n=5000)
q=6 (n=5000)
Density
f_u values
(b) Q-Q Plot (q=2)
Ordered Values
Theoretical quantiles
(c) Skewness vs q
Gaussian (0)
Skewness
q (biclique size)
(d) Excess Kurtosis vs q
Gaussian (0)
Excess Kurtosis
q (biclique size)
(e) Normality Test p-values
α=0.05
Jarque-Bera p-value
q (biclique size)
Mathematical Justification for Q3:
f_u = Σ_i X_i where X_i are independent noise terms
1. Central Limit Theorem:
• More terms (higher q) → CLT applies
• Sum of independent variables → Gaussian
2. Independence:
• RR perturbations: independent across edges
• Laplace noise: independent across queries
• Local counts: conditionally independent
3. Bounded Influence:
• Each term has bounded variance
• Lindeberg condition satisfied
• Higher-order cumulants decay as O(1/√n)
4. Empirical Evidence:
• Skewness → 0 as q increases
• Kurtosis → 0 as q increases
• Normality tests show improvement
Therefore: f_u ≈ N(μ, σ²) for q > 2